On finitariness of the Full-Ground Local State Monad

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1 Background

1.1 Full-Ground Local State

The monad for Full-Ground local state is as follows [Kam+17]:

$$\begin{aligned} \mathcal{T} \quad &: \quad [\mathbb{W}, \mathsf{Set}] \to [\mathbb{W}, \mathsf{Set}] \\ (\mathcal{T}X)w \quad &:= \quad \underbrace{\int_{w \rightarrowtail w' \in w/U} \mathbb{H}w' \to \overbrace{\int^{w' \rightarrowtail w'' \in w'/U} (XU)w'' \times \mathbb{H}w''}^{P(XU \times \mathbb{H})w'}}_{U_*(P(XU \times \mathbb{H})^{\mathbb{H}})w} \end{aligned}$$

The monad's action on objects can be described as taking the right Kan extension along the forgetful functor U of the functor $P(XU \times \mathbb{H})^{\mathbb{H}}$. In this case $P : [\mathbb{E}, \mathsf{Set}] \to [\mathbb{E}, \mathsf{Set}]$ is the so-called hiding monad and $\mathbb{H} : \mathbb{E} \to \mathsf{Set}$ represents heaps ($\mathbb{E}$ has objects worlds and morphisms as initialisation data).

$$\mathbb{E} \xrightarrow{\quad U \quad} \mathbb{W} \qquad\qquad [\mathbb{W}, \mathsf{Set}] \underset{U_*}{\overset{U^*}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} [\mathbb{E}, \mathsf{Set}] \circlearrowleft P$$

1.2 Finitary Monads

- A monad is said to be finitary if its underlying endofunctor preserves ω-filtered colimits.
- An ω-filtered colimit is a colimit over an ω-filtered category.
- An ω-filtered category is a category $\mathcal{C}$ that satisfies all of the following conditions:
  - There exists at least one object in $\mathcal{C}$.
  - For any two objects $C, D \in \mathcal{C}$, there exists an object E and two morphisms $f : C \to E$, $g : D \to E$.
  - For any two parallel morphisms $f, g : C \to D$, there exists a morphism $h : D \to E$ such that $h \circ f = h \circ g$.

2 Main Observation

Theorem 1. The Full-Ground Local State Monad is not finitary.

Proof. We will show that there exists a filtered category $\mathcal{C}$ and functor $F : \mathcal{C} \to [\mathbb{W}, \mathsf{Set}]$ such that

$$\mathcal{T}(\mathit{colimit}\, F) \ncong \mathit{colimit}\, (\mathcal{T}F)$$

in the category of cocones for the functor F.

Take $\mathcal{C} = \omega$, the posetal category where objects are natural numbers and morphisms are orderings $m \leq n$. This category is ω-filtered. Further define $F : \omega \to [\mathbb{W}, \mathsf{Set}]$ to be $Fn = \Delta_n$ that maps each natural number n to the constant functor mapping to the downset of n and $m \leq n$ are mapped to inclusions.

Notice that we can simplify *colimit* F:

$$(\mathit{colimit}\ F)w = \langle \biguplus_{n\in\mathbb{N}} n\downarrow \Big/ \sim, (i \in n\downarrow \mapsto q_n(i))_{n\in\mathbb{N}}\rangle \cong \langle \mathbb{N}, (i \in n\downarrow \mapsto i)_{n\in\mathbb{N}}\rangle$$

which extends to an isomorphism of cocones $\mathcal{T}(\mathit{colimit}\ F) \cong \langle \mathcal{T}(\Delta_{\mathbb{N}}), (\mathcal{T}(\Delta_n \hookrightarrow \Delta_{\mathbb{N}}))_{n\in\mathbb{N}}\rangle$.

We also know that:

$$\mathit{colimit}\,(\mathcal{T}F)w \cong \langle \biguplus_{n\in\mathbb{N}} \mathcal{T}\Delta_n w \Big/ \sim, (f \in \mathcal{T}\Delta_n w \mapsto q_n(f))_{n\in\mathbb{N}}\rangle$$

where the quotienting says that $\langle n, f \in \mathcal{T}(\Delta_n)w\rangle \sim \langle m, g \in \mathcal{T}(\Delta_m)w\rangle$ iff they are the same morphisms after we compose them with their corresponding inclusions to $\mathcal{T}(\Delta_{n+m})$.

The above observations boil down to showing that for some $w \in \mathbb{W}$:

$$\biguplus_{n\in\mathbb{N}} \mathcal{T}\Delta_n w \Big/ \sim \ncong \mathcal{T}(\Delta_{\mathbb{N}})w$$

where the mapping is given by $q_n(f \in \mathcal{T}(\Delta_n)w) \mapsto \mathcal{T}(\Delta_n \hookrightarrow \Delta_{\mathbb{N}})_w(f)$. We show that this mapping is not surjective.

Counterexample W.l.g. assume we only have one sort $\phi(C) = 1 + \mathsf{ref}_C$ and let $w = a : C$ be a singleton world. We define $\mathit{count} : \mathcal{T}(\Delta_{\mathbb{N}})w$ such that there cannot be element in $\mathcal{T}(\Delta_n)w$ for some n that represents it after post-composition with inclusion.

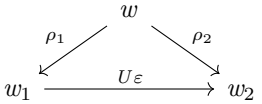
$$\begin{aligned}
&\mathit{count} : \int_{a:C\rightarrowtail w'\in a:C/U} \mathbb{H}w' \to \int^{w'\rightarrowtail w''\in w'/U} \mathbb{N}\times\mathbb{H}w'' \\
&\mathit{count}(\rho : a : C \rightarrowtail w')(h : \mathbb{H}w') = q_\rho\langle n, h\rangle \\
\text{where}& \\
&go = \mathsf{fix}\ \lambda rec.\, \lambda\langle a \in w', S \subseteq w'\rangle.\, \mathbf{case}\ \pi_a(h) \\
&\qquad\qquad\qquad\qquad\qquad\quad \mathbf{inl}\,().\,1 \\
&\qquad\qquad\qquad\qquad\qquad\quad \mathbf{inr}\ b.\ \begin{cases} 1 & \text{if } b \in S \\ 1 + rec(b, \{b\}\cup S) & \text{otherwise} \end{cases} \\
&n = go\langle a, \{a\}\rangle
\end{aligned}$$

Intuitively, *count* traverses the heap h starting from a and counts the number of different memory cells that it has encountered. Notice that *count* is defined for every extension of $a : C$ to w' and so there can be no finite $n\downarrow$ that can define it and so the mapping is not surjective. □

A note about the End in the FGLS monad The *count* function above is well-defined, i.e., it satisfies the End relation. To see why this is the case, notice that the End quantifies over all injections $\rho : w \to w'$, and thus it gives an ρ-ary product:

$$\{f_\rho : \mathbb{H}w' \to P(\Delta_{\mathbb{N}} \times \mathbb{H})w'\}_{\rho:w\to w'\in w/U}$$

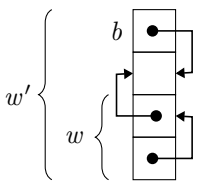
such that the following coherence condition holds:

For every initialisation of the form 

and heap $h : \mathbb{H}w_1$ we have:

$$P(\Delta_{\mathbb{N}} \times \mathbb{H})(\varepsilon)(f_{\rho_1}(h)) = f_{\rho_2}(\mathbb{H}\varepsilon h)$$

Intuitively, it says each component can use only the reachable part of a heap starting from locations in w. That is, take the following heap h where the location b is not reachable from locations in w.



The End relation says that the component f at world w' and heap h is completely determined by f at $w' \setminus \{b\}$ together with h without cell b. It is then clear to see that *count* satisfies the End relation as it traverses the heap *starting* from location a.

References

[Kam+17] Ohad Kammar, Paul Blain Levy, Sean K. Moss, and Sam Staton. "A monad for full ground reference cells". In: *LICS*. 2017. URL: https://doi.org/10.1109/LICS.2017.8005109.